

Anchored Graph Clustering with Local Feature Matching for AnimalCLEF 2026

Gilles Colling^{1,2}

¹*Division of BioInvasions, Global Change & Macroecology, Department of Botany and Biodiversity Research, University of Vienna, Vienna, Austria*

²*Vienna Doctoral School of Ecology and Evolution, University of Vienna, Vienna, Austria*

Abstract

AnimalCLEF 2026 evaluates individual animal discovery and re-identification across four species: Eurasian lynx, fire salamander, loggerhead sea turtle, and Texas horned lizard. Submissions assign test images to identity clusters and are evaluated using the Adjusted Rand Index (ARI) against hidden ground-truth identities. We present a per-species anchored graph-clustering framework combining pretrained re-identification descriptors (MiewID and MegaDescriptor), LightGlue local feature matching, and a tabular pairwise classifier for refinement edges. The system anchors the clustering graph to the labelled reference set: when a test image's blended similarity to a known training individual exceeds a species-specific threshold, it inherits that identity anchor. Test images sharing the same anchor are grouped together, while non-anchored pairs are merged only when supported by a high-confidence pairwise classifier score. Our submission, entered as team *gilles colling*, achieved 0.61741 ARI on the public leaderboard and 0.57038 ARI on the private leaderboard, placing 6th on the final (private) leaderboard.

Keywords

animal re-identification, individual identification, LightGlue, local feature matching, graph clustering, Adjusted Rand Index, AnimalCLEF, MiewID, MegaDescriptor

1. Introduction

Animal re-identification links a photograph to an individual animal. AnimalCLEF 2026 frames this as a *discovery* and clustering task: every test image is assigned to a cluster meant to represent one individual, and the test set may include individuals never seen during training, so the predicted cluster labels are themselves the discovery output rather than retrieval keys drawn from a closed training vocabulary [1, 2, 3, 4]. The official metric is the Adjusted Rand Index [10, 14], which compares the predicted clustering to the hidden identity ground truth after correcting for chance agreement. ARI penalises both directions of error: a false merge joins two animals and produces many wrong same-individual pairs, while a missed merge splits one animal across several clusters.

The competition spans four species with different individual-marking modalities: Eurasian lynx (coat patterns), fire salamander (dorsal spot patterns), loggerhead sea turtle (head and flipper scutes), and Texas horned lizard (ventral spot patterns) [5, 6]. Each species places different demands on the matching system: some are well-separated by global appearance descriptors, others require local feature matching to distinguish individuals on small markings.

We submitted a per-species graph-clustering pipeline. Pretrained re-identification backbones produce candidate similarity scores; LightGlue local matching produces a complementary score; species-specific thresholds turn high-confidence scores into edges in an undirected graph. Connected components of this graph become the predicted clusters. The pipeline anchors the graph through known training identities: for species with a labelled reference set, a test image whose nearest training neighbour exceeds the species threshold inherits that training individual's identity, and two test images mapped to the same training individual are merged automatically. This bridge produced many of the safe merges for the catalogue-rich species in the final submission.

CLEF 2026: Conference and Labs of the Evaluation Forum, September 21–24, 2026, Jena, Germany

✉ gilles.colling051@gmail.com (G. Colling)

ORCID 0000-0003-3070-6066 (G. Colling)



© 2026 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

The contributions of this paper are: (i) a description of a catalogue-anchored graph-clustering pipeline that scored 0.61741 ARI on the public leaderboard and 0.57038 ARI on the private leaderboard, with 833 clusters across 2 409 test images; (ii) a controlled component ablation on a train-as-test holdout showing that catalogue anchoring is the dominant component for the catalogue-rich species (Lynx, Sea Turtle), carried mainly by the global descriptors, while the pair-gate classifier rather than the anchor carries Salamander; (iii) reproducible per-species similarity blends and thresholds, with submitted-CSV SHA-256 hashes for traceability; and (iv) a documented set of failed experiments (head-crop matching, anatomy-based shape features, hierarchical agglomerative clustering, self-supervised backbone fine-tuning) to inform future submissions.

2. Related Work

2.1. Animal re-identification

Animal re-identification has shifted from species-specific hand-crafted pipelines to general-purpose deep metric learning [7, 9]. The WildlifeDatasets toolkit [7] aggregates many existing per-species datasets and provides a uniform evaluation framework; the associated MegaDescriptor model is a Swin-based backbone trained on this corpus across many species. WildlifeReID-10k [9] extends the same line with over 10 000 individuals across additional species, primarily as a pretraining corpus for general-purpose re-identification backbones. MiewID [8] is a multispecies re-identification model trained on a different community-curated corpus; it tends to generalise better than MegaDescriptor to unseen species, and the two models are complementary in practice.

2.2. Local feature matching

Local feature matching answers a different question than embedding similarity: whether visual points in two images can be put into geometric correspondence. LightGlue [13] is an attention-based matcher that operates on deep keypoint descriptors and that runs roughly an order of magnitude faster than its predecessor SuperGlue. Local matching is well-suited to patterned animals where individual identity is encoded in spots, scales, scutes, or markings rather than in overall body shape, and several recent re-identification systems combine local and global signals through calibrated fusion [7].

2.3. Open-set and clustering formulations

In a closed-set re-identification formulation, the system returns the most likely training identity for each query image. AnimalCLEF 2025 used an open-set variant in which each test image had to be classified either as a known training identity or as “new” [9], scored by a balanced known/unknown accuracy. AnimalCLEF 2026 changes the framing further: every test image must be assigned to a cluster, scored by ARI between the predicted and ground-truth partitions. This converts re-identification into a constrained graph-clustering problem in which global decisions about merges and splits matter as much as local pair decisions.

2.4. The Adjusted Rand Index

The Rand Index counts the fraction of item pairs that are placed consistently between two partitions of the same set, where consistent means grouped together in both partitions or kept apart in both. The Adjusted Rand Index [10] rescales this fraction so that a random labelling has expected score 0 and a perfect match has score 1. Given the contingency table between the predicted clusters $\{U_i\}$ and the ground-truth identities $\{V_j\}$, with $n_{ij} = |U_i \cap V_j|$, row sums $a_i = \sum_j n_{ij}$, column sums $b_j = \sum_i n_{ij}$, and n items in total,

$$\text{ARI} = \frac{\sum_{ij} \binom{n_{ij}}{2} - [\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}] / \binom{n}{2}}{\frac{1}{2} [\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}] - [\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}] / \binom{n}{2}}.$$

We use the scikit-learn implementation [14] for local cross-validation.

3. Task and Data

The official AnimalCLEF page links to the Kaggle competition that hosted the released data, sample submission, and leaderboard [1, 2, 4]. The lab page lists two headline species (Eurasian lynx and loggerhead sea turtle); the Kaggle release extends this to four species namespaces:

- LynxID2025 (946 test images),
- SalamanderID2025 (689 test images),
- SeaTurtleID2022 (500 test images); the namespace reuses the name of the SeaTurtleID2022 study [5], but the AnimalCLEF 2026 release is a different image set and should not be equated with the published SeaTurtleID2022 dataset,
- TexasHornedLizards (274 test images), linked to the Texas horned lizard study [6].

Each namespace ships with its own train split labelled with individual identities; the test split contains individuals that may or may not appear in the train split. Submissions are CSV files with columns `image_id` and `cluster`, where `cluster` is a label string scoped within the species namespace. Two test rows that share a cluster label are predicted to show the same individual.

The official metric is ARI, computed against the hidden ground-truth partition. Public-leaderboard scores are computed on a fraction of the test set during the competition; private (final) scores are computed on the complement and revealed at the end. Throughout this paper we report both: public-leaderboard ARI for development comparisons (where private scores were unavailable while we ran the experiments), and private-leaderboard ARI for the selected final submission.

Per-species visual cues. The four species place different demands on the matching system. Lynx individual identity lives in coat-spot patterns on the flanks and limbs; pose and lighting variation is large but coat patterns are reasonably stable across visits. Salamander individuals are separated by yellow dorsal spot patterns on a black background; the patterns are highly individual but image-to-image alignment is poor (zoom, body coverage, head-to-dorsal axis vary). Sea turtle individuals are identified from facial scute patterns visible in head shots [5]; aquatic backgrounds add water reflections and motion blur. Texas horned lizard individuals are identified from ventral spot patterns [6], with limited overlap between training and test photographs in our package, which makes test-test pair evidence the primary signal for this species.

4. System Overview

Figure 1 sketches the pipeline. The system runs independently per species: a lynx image is never compared to a salamander image. Within each species, three similarity signals are computed: global descriptor similarity, LightGlue local-matching similarity, and a learned pair-classifier score. These are combined into per-species similarity matrices, gated by species-specific thresholds, and resolved into connected components.

The submission build proceeds in seven steps:

1. Load metadata, image paths, the train/test split, and the labelled train identities.
2. Extract or load global descriptors from MiewID and MegaDescriptor for every image.
3. Extract or load LightGlue match scores for the train-test and test-test pairs that pass an initial global-similarity prefilter.
4. For Lynx, Salamander, and Sea Turtle, score every test image against every train identity using s_{final} and assign a train identity when the blended score exceeds τ_{anchor} .
5. For Horned Lizards, where the train-anchor signal is weak, build a test-test edge set directly from pair evidence.

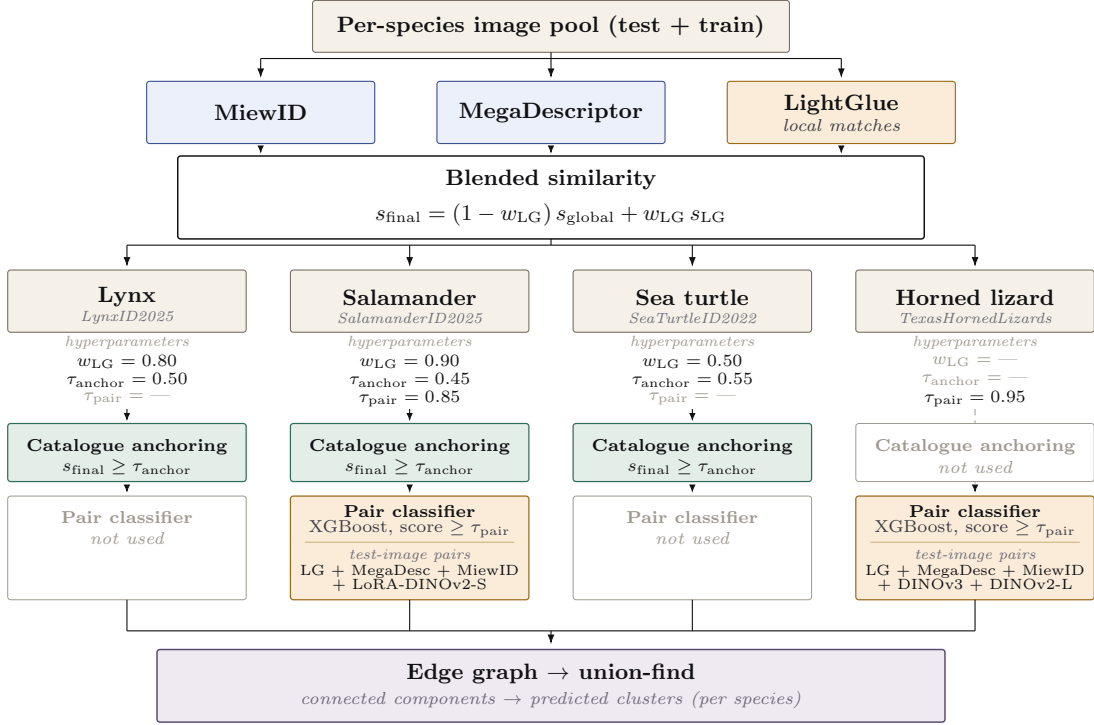


Figure 1: System overview. Per-species image pools are encoded with two pretrained global descriptors (MiewID, MegaDescriptor) and matched locally with LightGlue. The blended similarity s_{final} drives both catalogue anchoring (when a test image’s nearest training neighbour passes the species threshold τ_{anchor}) and the inputs to an XGBoost pair classifier that supplies refinement edges when its score passes τ_{pair} . Connected components of the resulting per-species graph become the predicted clusters.

6. Add conservative refinement edges from the learned pair classifier on top of the anchored graph for Salamander and Horned Lizard.
7. Resolve connected components with union-find and write the final CSV.

Table 1 gives the cluster shape of the best submitted run. Singleton fractions make the hard species visible: 52.2 % of Salamander and 85.0 % of Horned Lizard clusters are unmerged single images, against one of 36 for Lynx.

Table 1

Cluster shape of the selected final submission. Singletons are clusters of size 1; multi-image clusters are size ≥ 2 ; max is the largest cluster’s size.

Species	Rows	Clusters	Singletons	Multi-image	Max
LynxID2025	946	36	1 (2.8%)	35	156
SalamanderID2025	689	402	210 (52.2%)	192	25
SeaTurtleID2022	500	169	48 (28.4%)	121	11
TexasHornedLizards	274	226	192 (85.0%)	34	7
Total	2409	833	451 (54.1%)	382	—

5. Visual Similarity Signals

5.1. Global descriptors

MiewID [8] and MegaDescriptor [7] produce L_2 -normalised vector embeddings per image. For two images, cosine similarity between their vectors gives a whole-image identity score. The submitted

system uses a fixed weighted blend

$$s_{\text{global}} = 0.7 s_{\text{MiewID}} + 0.3 s_{\text{MegaDescriptor}},$$

tuned on local cross-validation. The global signal is fast enough to score every train-test pair across all species, and serves as the prefilter that decides which pairs are scored with LightGlue.

5.2. LightGlue local matching

LightGlue [13] matches deep keypoints (SuperPoint [18] front-end) between two images and returns a count of accepted matches after geometric verification. Counts are normalised per species by a rank transform: each pair’s raw match count is mapped to its empirical percentile over all scored pairs of that species, giving a similarity score $s_{\text{LG}} \in [0, 1]$ that is robust to the heavy-tailed match-count distribution. The blended similarity used in catalogue anchoring is

$$s_{\text{final}} = (1 - w_{\text{LG}}) s_{\text{global}} + w_{\text{LG}} s_{\text{LG}},$$

with the per-species LightGlue weight w_{LG} and anchoring threshold τ_{anchor} given in Table 2.

5.3. Learned pair scores

Salamander and Horned Lizard each use a separate gradient-boosted tree model (XGBoost [11]) trained on tabular features per image pair. Both classifiers share a common base of features: raw, rank-normalised, and percentile-ranked LightGlue counts, MegaDescriptor and MiewID cosine similarities, and a co-clustering prior tuned on the public leaderboard. The co-clustering prior is a binary feature indicating whether the two images shared a cluster in an earlier submission whose per-species thresholds were tuned on the public leaderboard; because that earlier clustering was itself fit to the public split, this feature can carry public-split-specific information, which contributes to the public-to-private gap discussed in Section 10. The species-specific features differ in what global descriptor evidence they add on top of this base: the Salamander classifier adds a LoRA-fine-tuned [12] DINOv2-S [15] cosine that targets dorsal spot identity, while the Horned Lizard classifier adds DINOv3 [16] (CLS, patch-max, patch-mean, attention-pooled) and DINOv2-L cosines, which are more useful when the local-matching signal is weak. Each classifier is trained on manually labelled image pairs. We inspected images of the target species and recorded same/different-individual verdicts on test–test and test–train pairs; for the Salamander spot-matching adapter (Section 12) we additionally marked corresponding spot regions between same-individual images. The Salamander classifier is trained on 165 confirmed same-individual pairs (8 test–test, 157 test–train) within 993 labelled Salamander verdicts; the Horned Lizard classifier on 10 confirmed test–test pairs within 617 verdicts. The trained model scores a pair from its image features and is applied to pairs it never saw labelled; both the scores and their effect on the submitted clustering are reproducible from the released code and weights (Section 12). We use the classifier only as a refinement gate: only pairs whose classifier score exceeds the species-specific gate τ_{pair} contribute edges to the graph.

5.4. Hyperparameters

Table 2 collects the per-species hyperparameters of the submitted system. The LightGlue weight w_{LG} controls how aggressively local matching dominates the blended similarity; τ_{anchor} is the catalogue-anchor threshold; τ_{pair} gates the pair classifier output. Where τ_{pair} is not listed, the species path does not use the pair classifier as a refinement gate.

Table 2

Per-species hyperparameters of the submitted system.

Species	w_{LG}	τ_{anchor}	τ_{pair}
LynxID2025	0.80	0.50	—
SalamanderID2025	0.90	0.45	0.85
SeaTurtleID2022	0.50	0.55	—
TexasHornedLizards	—	—	0.95

6. Species-Specific Pipelines

6.1. LynxID2025

Lynx contributes the largest test set (946 images). Coat patterns are stable across visits, so the global descriptor blend already separates individuals well; LightGlue mainly helps when the global descriptor cannot distinguish two visually similar individuals. With $w_{LG} = 0.80$ and $\tau_{\text{anchor}} = 0.50$, the submitted clustering anchors most test images to a small set of training identities (Table 1). Single-species experiments, including triplet fine-tuning of a DINOv2 [15] backbone on the Lynx train split, did not improve leaderboard ARI, so we kept the descriptor-based path that the leaderboard had already validated.

6.2. SalamanderID2025

Salamander placed the greatest demands on the matching system. Dorsal spot patterns are highly individual, but image-to-image alignment is poor: the same individual can appear at very different zoom levels, with different parts of the body in frame, against different backgrounds. Local matching produces many candidate edges; many visually plausible ones are false positives near the clustering boundary.

The submitted Salamander path uses a LightGlue-heavy blend ($w_{LG} = 0.90$, $\tau_{\text{anchor}} = 0.45$) for catalogue anchoring, then adds Salamander test-test merges from the learned pair classifier gated by the pair threshold τ_{pair} . The submitted setting $\tau_{\text{pair}} = 0.85$ was selected by local cross-validation against train-as-test holdout merges. Lower values added too many unsafe edges; higher values missed useful merges. The complete Salamander procedure is encoded in the released code (Section 12).

6.3. SeaTurtleID2022

Sea turtle scute patterns are well-captured by the global descriptor blend, so this species uses a balanced $w_{LG} = 0.50$ with $\tau_{\text{anchor}} = 0.55$ and no learned-pair refinement gate. Early leaderboard runs showed that broad Sea Turtle changes which improved local pair metrics often reduced overall ARI, so the species path was kept conservative for the rest of the competition.

6.4. TexasHornedLizards

The Horned Lizard split has limited train-test overlap: many test images depict individuals with no clear nearest training neighbour. The submitted Horned Lizard path therefore relies on test-test pair evidence. The learned pair classifier is applied to all test-test pairs and edges are kept only when the score exceeds the high pair threshold $\tau_{\text{pair}} = 0.95$, leaving an intentionally conservative clustering with many singleton components (Table 1). A hierarchical agglomerative clustering (HAC) variant on the same pair-score matrix scored 0.57706 on the public leaderboard (Table 4) and was rejected in favour of this graph-merging baseline.

7. Catalogue Anchoring

Catalogue anchoring connects the clustering graph to the labelled reference set. For a test image x , the system searches train images of the same species, computes s_{final} against each, and selects the best train neighbour. If s_{final} exceeds the species threshold τ_{anchor} , x inherits that train individual’s identity. Two test images that inherit the same train identity are automatically merged.

This changes the graph problem in two ways (Figure 2). First, a test-test-only method has to prove that two test images match each other directly; anchoring lets the system route them through a shared reference identity instead, which is often easier when poses, crops, or lighting differ but both images match the same known animal. Second, the anchor labels are deterministic across submissions, so iterative refinement can preserve the anchored backbone and only experiment with test-test merges on top of it.

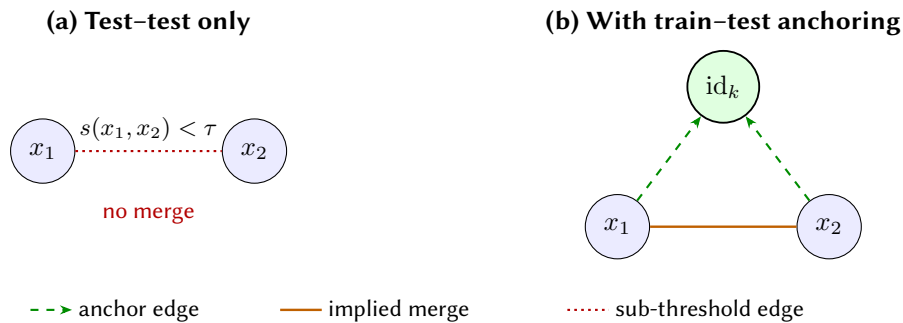


Figure 2: Train–test anchoring. (a) A direct test–test method must score the pair (x_1, x_2) above the species threshold; when poses, crops, or lighting differ, the direct score $s(x_1, x_2)$ often falls short. (b) If both x_1 and x_2 separately match a known training individual id_k above τ_{anchor} , they inherit its label and are merged automatically, recovering pairs that direct test–test scoring misses.

The follow-up rows in Table 4 make this concrete: every variant that added or rescored test-test edges without preserving the anchored backbone scored below the anchored run on the public leaderboard.

8. Component Ablation

The leaderboard comparisons in Table 4 vary several components at once. To isolate each component’s contribution we ran a controlled ablation on a train-as-test holdout, the only validation surface available after the leaderboard closed. Each species’ labelled training set is split into an open-set holdout (5 folds, seed 42): 30% of identities are held out entirely as queries with no catalogue entry, and each remaining identity keeps one catalogue reference and contributes its other images as queries, so the anchor has multiple query images per catalogued individual to join. The Adjusted Rand Index is computed on the query partition against the known identities. The submitted anchor thresholds are calibrated on the test-to-train similarity scale and do not transfer to this train-to-train surface, so we tune each configuration’s anchor threshold on the holdout by a keep-fraction sweep and report its best mean ARI. Absolute ARI on this surface is not comparable to the leaderboard; the differences between configurations are the result.

The catalogue anchor is the dominant component for the two catalogue-rich species (Table 3). For Sea Turtle it raises ARI from 0.084 to 0.625, and for Lynx from 0.019 to 0.149, in both cases well above any test-test clustering of the same images. For these species the anchor is carried mainly by the global descriptors: anchoring on the global descriptors alone matches the full blend to within 0.004, while anchoring on LightGlue alone is much weaker (0.300 and 0.038). A threshold-free check agrees: the top-1 catalogue accuracy (the fraction of closed queries whose nearest catalogue neighbour has the correct identity) is 0.726 for Sea Turtle and 0.232 for Lynx under the blend, with the blend at or above either single signal for all three species.

Table 3

Controlled component ablation on the train-as-test holdout (mean ARI over 5 folds; per-configuration anchor threshold tuned on the holdout). No anchor: the test-test mutual-kNN and average-linkage clustering on the blended similarity. Anchor (blend / global / LightGlue): catalogue anchoring driven by the full blend, the global descriptors alone, or LightGlue alone.

Species	No anchor	Anchor (blend)	Anchor (global)	Anchor (LightGlue)
LynxID2025	0.019	0.149	0.153	0.038
SeaTurtleID2022	0.084	0.625	0.622	0.300
SalamanderID2025	0.234	0.122	0.106	0.125

The Salamander result departs from this pattern. A direct test-test clustering reaches 0.234 ARI here, above the 0.122 of the anchored pipeline: catalogue anchoring does not help this species on the holdout, where catalogue references are sparse (about 2.4 images per identity) and top-1 catalogue accuracy is low (0.225). The component that carries Salamander is instead the pair-gate classifier, which adds 0.086 ARI over the anchor alone (0.037 to 0.122); and in contrast to the other two species, LightGlue rather than the global descriptors is the stronger anchor signal for Salamander (top-1 0.219 against 0.150).

These results refine the leaderboard reading: catalogue anchoring is the dominant component for the catalogue-rich species (Lynx, Sea Turtle) and is best driven by the global descriptors, whereas Salamander depends on the pair-gate classifier and on LightGlue.

9. Failed Experiments

We document four directions that improved local pair-level diagnostics without translating into leaderboard ARI gains, to inform future submissions.

Head-crop matching for Salamander. We trained a Salamander head detector and ran LightGlue matching only inside the detected head crop, hypothesising that head spots are more discriminative than full-dorsal spots. Local pair AUC on labelled head-crop pairs was high, but the resulting head-crop similarity matrix did not change the public leaderboard score: the head-crop edges were largely redundant with edges already supplied by full-image LightGlue.

Anatomy-based shape features for Salamander. We computed dorsal-spine curvature and body-silhouette descriptors on segmented Salamander masks. These features had non-trivial pair AUC but contributed almost no new high-confidence edges to the graph.

Self-supervised Salamander backbone. We fine-tuned a DINOv2 [15] backbone with self-supervised pretraining on the Salamander train split, expecting more discriminative embeddings than MiewID for this species. Train-train pair AUC improved, but the resulting embeddings did not produce safe new catalogue-anchor merges at any threshold tested.

Test-test Salamander redesign. A late-stage redesign rescored Salamander test-test edges from scratch and dropped the catalogue anchor. It scored 0.59207 on the public leaderboard, well below the anchored 0.61741.

All four experiments improved pair-level diagnostics (AUC, top-1% precision, hard-edge accuracy) without changing leaderboard ARI. Pair-level gains only translate into ARI if the new edges survive the operating threshold the merge graph actually uses and add coverage on top of the anchored backbone instead of replacing it.

10. Submitted Runs

Table 4 lists the main leaderboard submissions, identified by the SHA-256 prefix (8 hex characters) of the submitted CSV. Private-leaderboard scores are reported only for the selected final submission, because Kaggle reveals private scores only for entries the team selects as final.

Table 4

Leaderboard submissions, grouped by method. The “Anchored graph (final)” row was selected as the team’s final submission and is the only row with a recorded private-LB score.

Method family	Description	Public LB	Private LB	SHA-8
LightGlue baseline	Species-specific local-matching and descriptor baseline	0.56792	—	3CC406B7
Pair-refined baseline	Baseline with learned pair refinements	0.59644	—	58F410BD
Anchored graph (final)	Catalogue anchoring with conservative Salamander refinement	0.61741	0.57038	C772B87C
Alternative pair-gate setting	Pair-gate variant with a different threshold	0.61259	—	69C8754D
Additive Salamander edge model	Extra Salamander test-test edges added to the anchored graph	0.60156	—	DF2ADD2C
Test-test Salamander redesign	Rebuilt Salamander edges without preserving the catalogue anchor	0.59207	—	E4C31AB8
Horned Lizard HAC probe	Agglomerative-clustering probe for Horned Lizard	0.57706	—	C2048236

Full SHA-256 hashes for the submitted CSVs:

```
3CC406B7C75A82C8122395875A4040AD7D0B9265F70514F522EE92169067F1D4
58F410BD9C0E374D2A57FDAC1B6A3D95006EBF2746BF01B192278CE3B5033A3A
C772B87C7989B0C23D360EC895934236ED48D1D9C055C2DE0C010833E6598DF6
69C8754D983C43FDCA811F792D3F68B58875D93E20FA1335A8669B1964FEA92E
DF2ADD2C16FE48327CE76A3677EA223654419784C6002B22B4E395D66A57AE8B
E4C31AB87E89A90027B26B607ABD7594D271153F665147C394592B926F56DEF9
C2048236EF0C929885F17D20C7D2503101D715D019DB4532994515430121A6E1
```

Reproducibility details. LightGlue is computed only for image pairs that survive a global-similarity prefilter (the high- s_{global} train-test and test-test neighbours of each image), which bounds the otherwise quadratic matching cost. Classifier training pairs are built from hand-confirmed same/different-individual labels with leave-image-out folds ($K=10$), so no image appears on both sides of a fold. The per-species thresholds of Table 2 were searched by directly maximising public-leaderboard ARI—each row of Table 4 is one such probe—with the Salamander pair gate additionally checked on the train-as-test holdout. All descriptor extraction, LightGlue matching, classifier training, and clustering ran on a single Apple M4 Pro; runtime was dominated by one-time descriptor and LightGlue extraction and was not separately profiled. Table 5 summarises which validation surface each reported number comes from.

Table 5

Validation surfaces and the role of each. Public-leaderboard ARI is fit to its split (it selects the operating point); the private leaderboard is the only held-out evaluation.

Surface	Role	Final-run ARI
Train-as-test holdout	Component ablation and threshold sanity (Table 3); different distribution from the test set	Table 3
Public leaderboard	Operating-point selection; thresholds fit to this split	0.61741
Private leaderboard	Final held-out evaluation, revealed at close	0.57038

10.1. Public and Private Scores

The selected submission, entered as team *gilles colling*, scored 0.61741 ARI on the public split and 0.57038 ARI on the private split (6th on the final leaderboard), a difference of 0.046. The per-species thresholds in Table 2 were chosen by maximising public-LB ARI directly: each entry in Table 4 corresponds to a public-LB probe used to update one or more thresholds. Tuning the operating point against the public split is itself a form of overfitting to that split, and the 0.046 drop is the gap between the two ARI estimates this procedure produces. We have no separate held-out estimate of generalisation error.

Use of test-set annotations. The pair classifiers and the Salamander spot adapter are trained on manual annotations of the competition test images: same/different verdicts on test-test and test-train pairs, and marked spot regions between same-individual images (counts in Section 12). These are used as training labels; the trained models score every candidate test pair through the same threshold, and the clustering is reproducible from the released code and weights.

11. Limitations

The submitted system has five known limitations. The catalogue anchor relies on test images of training individuals existing in the test set; for species or splits where this overlap is small (Horned Lizard in our package), the anchor signal is weak and the system falls back to less reliable test-test pair evidence. The per-species thresholds were tuned on a combination of public-leaderboard feedback and local train-as-test cross-validation; they may not transfer to the private leaderboard or to a different release. The learned pair classifier operates on descriptor cosine similarities and LightGlue match counts only; it has no access to image content beyond these summaries, which limits its ability to disambiguate individuals when both signals are similar. The controlled ablation in Section 8 runs on a train-as-test holdout, a different distribution from the leaderboard test set; its absolute ARI is not leaderboard-comparable, so only the per-configuration differences are interpretable, and the closed leaderboard precludes a held-out check. Finally, the LoRA-DINOv2-S adapter for Salamander was trained on a hand-labelled set of approximately 7 800 keypoint correspondences (Section 12); the trained adapter weights are released, but this annotation set would be laborious to reconstruct from scratch, which limits independent re-training of that one component.

12. Code and Data Availability

The released AnimalCLEF 2026 dataset, sample submission, and leaderboard are available through the official Kaggle competition page [1]. The submitted system has the following trained components. (i) MegaDescriptor [7] (BVRA/MegaDescriptor-L-384) was fine-tuned with ArcFace [17] heads on the AnimalCLEF 2026 train split for Lynx and Sea Turtle on Apple M4 Pro (Metal); these checkpoints supply the species-specific MegaDescriptor embeddings used in the similarity step. (ii) MiewID [8] (conservationxlabs/miewid-msv3) was used unchanged as a pretrained encoder; the `cos_miew` feature seen by the pair classifiers comes from the upstream MiewID embeddings. (iii) For the Salamander branch, a LoRA adapter (rank 8, $\alpha = 16$, on query and value projections) was trained on DINOv2-S [15] over a hand-labelled Salamander spot-correspondence dataset of approximately 7 800 colour-coded keypoint correspondences across image pairs of the same individual. The objective is a CLIP-style symmetric InfoNCE loss that pulls together 128×128 patch crops at matching keypoints and pushes apart patches at different keypoints on the same image pair, so that the encoder discriminates spot-level identity rather than generic "yellow-blob" texture. The resulting pair-similarity matrix is blended with LightGlue in the Salamander clustering step, and the LoRA-DINOv2-S cosine is one of the input features to the Salamander pair classifier. (iv) Two XGBoost [11] pair classifiers (one for Salamander, one for Horned Lizard) were trained leave-image-out ($K=10$) on the manually labelled same/different pairs, and score every test-test pair at inference time; pairs scoring at or above the species τ_{pair} contribute

edges to the clustering graph. Both classifiers use 300 trees of depth 3, learning rate 0.05, subsample and column-subsample 0.8, L_2 regularisation 1.0, and class-balanced positive weighting (binary:logistic objective, histogram tree method). (v) LightGlue [13] with the SuperPoint [18] front-end was used unchanged. Submitted CSVs are uniquely identified by the SHA-256 hashes listed in Section 10.

Pipeline source code, per-species YAML configs encoding the hyperparameters of Table 2, the submission SHA-256 manifest, and a verifier script are released at <https://github.com/gcol33/animal-clef-2026> under an MIT licence. The trained weights for the leaderboard run – two MegaDescriptor fine-tunes (Lynx, Sea Turtle), the LoRA-DINOv2-S adapter and its spot detector for Salamander, and the two XGBoost pair classifiers (Salamander and Horned Lizard) – are released separately on the Hugging Face Hub at <https://huggingface.co/gcol33/animal-clef-2026> under CC-BY-4.0; the MANIFEST.md in that repository documents each file’s role and source, and the GitHub README pins the commit revision used for the submitted runs. Image data is not redistributed and must be obtained from the Kaggle competition page.

Licences and dataset acknowledgments. Our code release is MIT-licensed and our trained weights are released under CC-BY-4.0, subject to the upstream restrictions noted here. The Sea Turtle MegaDescriptor fine-tune is a derivative of training on the AnimalCLEF 2026 Sea Turtle split, which is sourced from SeaTurtleID2022 [5]; SeaTurtleID2022 is released for non-commercial research use only and may not be re-uploaded, so the released Sea Turtle weights may not be used for commercial purposes. Backbones used unchanged or as fine-tuning starting points (BVRA/MegaDescriptor-L-384, conservationlabs/miewid-msv3, facebook/dinov2-small, and the DINOv3 checkpoints used in the Horned Lizard branch) retain their upstream licences. The AnimalCLEF 2026 train/test split itself is governed by the Kaggle competition terms [1]. No raw images, JPEG bytes, or dataset paths from any upstream source are redistributed in either the GitHub repository or its Hugging Face mirror.

13. Conclusion

Our best AnimalCLEF 2026 submission scored 0.61741 ARI on the public leaderboard and 0.57038 ARI on the private leaderboard, with a per-species graph-clustering pipeline. Pretrained MiewID and MegaDescriptor descriptors supplied broad identity similarity, LightGlue local matching supplied complementary pair evidence, and species-specific thresholds controlled when each signal contributed graph edges. The controlled ablation (Section 8) locates this advantage: catalogue anchoring is the dominant component for the catalogue-rich species, Lynx and Sea Turtle, where routing test images through known training identities far exceeds clustering the same images directly, and where the anchor is driven mainly by the global descriptors. For Salamander, the pair-gate classifier rather than the anchor carries the clustering.

We therefore suggest future iterations keep the anchor as the backbone for catalogue-rich species and invest in the test-test signal (the pair-gate classifier and local matching) for sparse-catalogue species such as Salamander; new neural or anatomy-based signals earn their place by adding safe edges to the full all-species graph, evaluated at the clustering level. Pair-level diagnostics (AUC, top-1% precision, hard-edge accuracy) were useful for debugging in this work; the four runs in Section 9 show that strong pair-level numbers can leave leaderboard ARI unchanged.

Acknowledgments

We thank the AnimalCLEF organisers for releasing the data and running the leaderboard, and the maintainers of MiewID, MegaDescriptor, and LightGlue for making the pretrained models openly available.

Declaration on Generative AI

During the development of the system, the author used a local REAP-48B model for coding assistance (typing speedup and routine code transformations); REAP-48B is a 48-billion-parameter sparse mixture-of-experts checkpoint derived from Qwen3-Next-80B [20] via the REAP expert-pruning method [19], served on an Apple M4 Pro. No proprietary or remote LLM was used. The machine-learning design and the prose of this manuscript are the author’s own work; the author takes full responsibility for the publication’s content.

References

- [1] L. Adam, L. Pícek, K. Papafitsoros, D. Williams, and D. Biffi. AnimalCLEF26 @ CVPR & CLEF. Kaggle, 2026. <https://www.kaggle.com/competitions/animal-clef-2026>
- [2] ImageCLEF. AnimalCLEF2026: Discovery and Re-Identification of Individual Animals. <https://www.imageclef.org/AnimalCLEF2026>
- [3] L. Pícek, L. Adam, S. Kahl, R. Bossy, L. Chrobak, H. Goëau, K. Papafitsoros, H. Klinck, W.-P. Vellinga, R. Planqué, T. Denton, K. Barnard, C. Nédellec, L. Deléger, M. Courtin, G. Martellucci, I. Moummad, F. Vinatier, P. Bonnet, and A. Joly. Overview of LifeCLEF 2026: AI challenges for biodiversity understanding and ecosystem management. In *International Conference of the Cross-Language Evaluation Forum for European Languages (CLEF)*. Springer, 2026.
- [4] L. Adam, K. Papafitsoros, D. A. Williams, D. Biffi, and L. Pícek. Overview of AnimalCLEF 2026: Discovery and re-identification of individual animals. In *Working Notes of CLEF 2026 – Conference and Labs of the Evaluation Forum*, 2026.
- [5] L. Adam, V. Čermák, K. Papafitsoros, and L. Pícek. SeaTurtleID2022: A long-span dataset for reliable sea turtle re-identification. In *Proc. IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 7131–7141, 2024. <https://doi.org/10.1109/WACV57701.2024.00699>
- [6] D. Biffi, M. R. Tucker, A. Ackel, and D. A. Williams. Identification of individual Texas Horned Lizards (*Phrynosoma cornutum*) using genotypes and ventral spot patterns. *Ecology and Evolution*, 15(3):e71167, 2025. <https://doi.org/10.1002/ece3.71167>
- [7] V. Čermák, L. Pícek, L. Adam, and K. Papafitsoros. WildlifeDatasets: An open-source toolkit for animal re-identification. In *Proc. IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 5941–5951, 2024. <https://doi.org/10.1109/WACV57701.2024.00585>
- [8] L. Otarashvili, T. Subramanian, J. Holmberg, J. J. Levenson, and C. V. Stewart. Multispecies animal re-ID using a large community-curated dataset. *arXiv preprint arXiv:2412.05602*, 2024. <https://doi.org/10.48550/arXiv.2412.05602>
- [9] L. Adam, V. Čermák, K. Papafitsoros, and L. Pícek. WildlifeReID-10k: Wildlife re-identification dataset with 10k individual animals. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 2124–2134, 2025. <https://arxiv.org/abs/2406.09211>
- [10] L. Hubert and P. Arabie. Comparing partitions. *Journal of Classification*, 2:193–218, 1985. <https://doi.org/10.1007/BF01908075>
- [11] T. Chen and C. Guestrin. XGBoost: A scalable tree boosting system. In *Proc. 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. <https://doi.org/10.1145/2939672.2939785>
- [12] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- [13] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys. LightGlue: Local feature matching at light speed. In *Proc. IEEE/CVF International Conference on Computer Vision*, pages 17581–17592, 2023. <https://doi.org/10.1109/ICCV51070.2023.01616>
- [14] scikit-learn developers. `sklearn.metrics.adjusted_rand_score`. https://scikit-learn.org/stable/modules/generated/sklearn.metrics.adjusted_rand_score.html
- [15] M. Oquab, T. Darcet, T. Moutakanni, H. V. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.
- [16] O. Siméoni, H. V. Vo, M. Seitzer, F. Baldassarre, M. Oquab, C. Jose, V. Khalidov, M. Szafraniec, S. Yi, M. Ramamonjisoa, et al. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025. <https://doi.org/10.48550/arXiv.2508.10104>
- [17] J. Deng, J. Guo, N. Xue, and S. Zafeiriou. ArcFace: Additive angular margin loss for deep face recognition. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4690–4699, 2019. <https://doi.org/10.1109/CVPR.2019.00482>
- [18] D. DeTone, T. Malisiewicz, and A. Rabinovich. SuperPoint: Self-supervised interest point detection and description. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 224–236, 2018. <https://doi.org/10.1109/CVPRW.2018.00060>
- [19] M. Lasby, I. Lazarevich, N. Sinnadurai, S. Lie, Y. Ioannou, and V. Thangarasa. REAP the experts: Why pruning prevails for one-shot MoE compression. *arXiv preprint arXiv:2510.13999*, 2025. <https://doi.org/10.48550/arXiv.2510.13999>
- [20] A. Yang, A. Li, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Gao, C. Huang, C. Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. <https://doi.org/10.48550/arXiv.2505.09388>